

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			Correct substitution and rearrangement into the speed formula (1) $t = \frac{7.2 \times 10^7}{3 \times 10^8} = 0.24 \text{ [s]} \text{ (1)}$ $t = \frac{6.0 \times 10^6}{2 \times 10^8} = 0.03 \text{ [s]} \text{ (1)}$ So disagree Conclusion must be present to award all 3 marks Alternative: Speed of the signal to the satellite is 1.5 times faster than in the cable (1) The distance travelled to the satellite and back is 12 times greater than the cable (2) So disagree Conclusion must be present to award all 3 marks Award a maximum of 2 marks for answers of 0.48 [s] and 0.06 [s]			3	3	3	
				Question 4 total	3	2	3	8	3	0